

society as a whole because they promote social solidarity, i.e., the concept of sharing and cooperation.”

There is a vast difference between ‘software which you can get for zero cost’ and ‘software which gives you the freedom to use it the way you want to’. You cannot look into the source code of free software (zero cost software or pirated software that is distributed freely), but you have access to the source code of open source software.

Open sourcing software has its own advantages. From a user’s perspective, the obvious advantage is the software being freely available; for developers or programmers, the benefit is getting access to the source code to do whatever they want.

Another great advantage of OSS is that the software continuously evolves, as more and more developers contribute to it, adding to and modifying it, which makes the software better, more secure and relatively bug-free, compared to proprietary software. The best example is the Linux kernel. The rate of development of the Linux kernel is unmatched. Some recent statistics announced by LinuxFoundation.org highlight this aspect: “Nearly 12,000 developers from more than 1,200 companies have contributed to the Linux kernel since tracking began 10 years ago. Just since the last report, more than 4,000 developers from 200 companies have contributed to the kernel, half of whom contributed for the first time. The average number of changes accepted into the kernel per hour is 7.71, which translates to 185 changes every day and nearly 1,300 per week.”

Today, big names like Google, Facebook, Intel, Samsung, Red Hat, Canonical, Cisco, Yahoo, etc., are promoting and contributing to open source activities.

The need for open source

It all started with the frustration of not being able to tweak the software we used. In the early 1980s, Richard Stallman, a computer programmer and hacker, and a few others, were not allowed to modify the code of a newly installed laser printer at AI Labs, where they worked. Stallman had modified the source code of the lab’s previously installed printer so that it sent electronic notifications to the user when its printing job was completed. Stallman wanted to add the same functionality to the newly installed printer but was prevented from doing so. This and a few other similar events triggered the birth of open source software.

Today, in this rapidly developing digital era, open source software plays a very important role. Well known OSS include the Linux kernel; the well-known browser, Mozilla; the Apache Web server that powers most of the world’s websites; OpenSSL, a project that keeps the Internet secure and is used by most organisations and governments; GNUPG, which is encryption software used by most organisations to secure mails and files; and NTP (Network Time Protocol), which synchronises the time of all machines over the Internet. There is also the very well known and widely used open cloud software, OpenStack... and the list goes on and on!

Organisations like the Linux Foundation, which supports the development of the Linux kernel as well as other open source projects; and the Apache Software Foundation, which backs the development of the Apache Web server software and other such projects, are examples of how successful and widespread the OSS ideology has become.

The power of collaboration that the OSS ideology has unleashed is best demonstrated by the Linux kernel, which is developed and maintained by the community. The reason the Linux kernel is being used in the palm-sized Raspberry Pi computers, in supercomputers that power space stations, in cars as well as submarines that dive deep into the sea, is because people from around the world collaborate and add patches to the kernel, which enables it to support such diverse hardware.

If Linus Torvalds, the creator of Linux, had felt like keeping his project to himself, he would have ended up founding another Microsoft, and the world as we know it would have never been the same. So definitely, the future is open!

How open source works

Let’s assume you find some open source software and start using it. Then you stumble across a bug, or you would like to add a new feature to it, so you contact the development team. If it’s a bug you’ve found, you submit it to a bug tracker. If it’s a new feature or patch that you’d like to add, and if the team likes your idea, you might be asked to write the patch yourself!. If it is an easy task, then you can yourself modify the code, do some tests and submit the patch. If the team accepts the patch and applies it, then you are happy that your contribution makes the software even better.

But what if you’ve created amazing software and want to open source it. Again, it is simple. You create a zip of your code and publish it on one of the open source software hosting sites like GitHub at github.com, SourceForge at sourceforge.net, Google Code at code.google.com, Gitorious at gitorious.org, and many more. Once your project is out there, people will come across it, and start collaborating.

So the development of OSS happens collaboratively.

So who’s building products on top of OSS?

The answer is - almost everyone! The tech giant, Google, has contributed over 20 million lines of code from over 900 open source projects. The best example is Android, which is a software stack for mobile devices that is based on Linux. Chromium is Google’s Web browser, Ganeti is a cluster virtual server management software, Gerrit is a Web based code review system, Go is a programming language, apart from many more.

But Google isn’t the only big firm in this space. Its rival, Facebook, is also in the race. Even Facebook has a wide range of open source projects that cover Web as well as back-end servers. Buck is a build system for Android that helps in building reusable modules, Bolts comprises a few libraries for